

CPS 630 - Web Applications

Project-It1-It2-W25

Team 12

Archan Joshi (500950854)

María Fernanda Ceballos Gonzalez (501122751)

Student Name and Number	Tasks Actioned	Percentage of Contribution on whole project
Archan Joshi (500950854)	Design and implementation of sign-up/log-in functions, database, implementing MD5 hashing function, writing the final report	50%
Maria Fernanda Ceballos (501122751)	Design and implementation of the UIs, coding and debugging connection with server-side, writing the final report	50%

Project Objectives and Technology Used

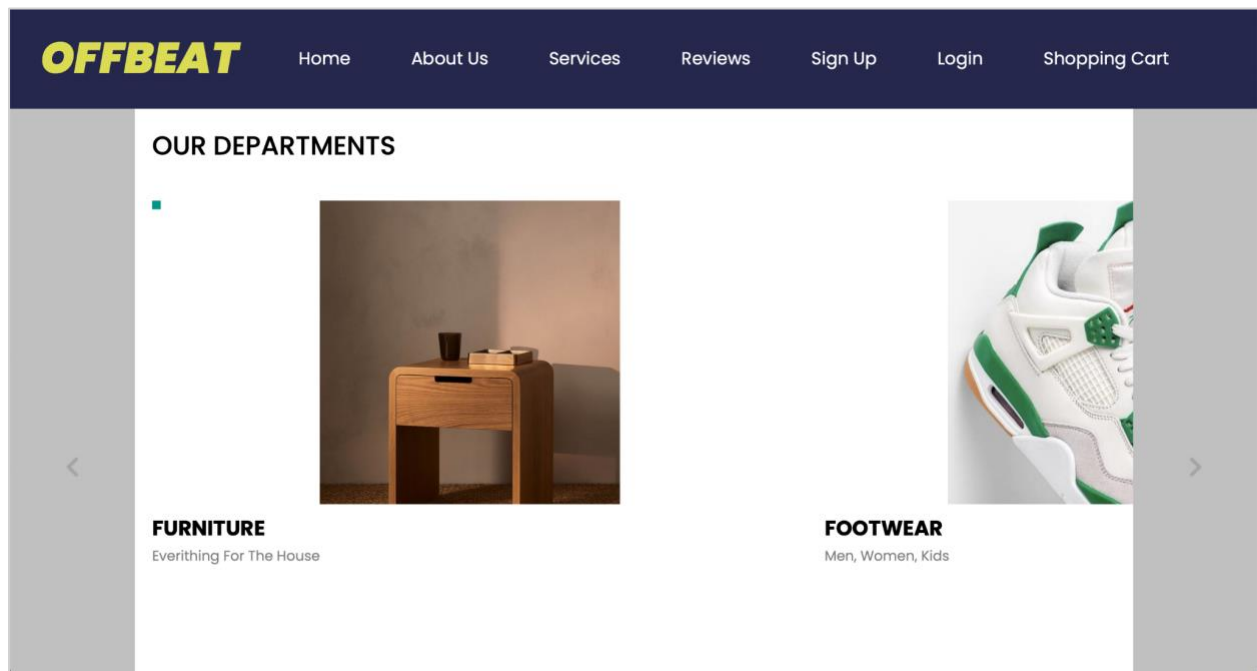
In this project, we were tasked in improving upon our Online Service Platform (OSP) that could connect multiple web pages in order to create a platform that handles shopping, delivery, accepting payments, and user sign-up and login. Our specific design refers to the website called “OFFBEAT”, an online sneaker retailer that houses warehouses in both Canada and the United States. The key improvements made were:

1. Re-designing the webpage to function as a Single-Page-Application (SPA)
2. Implementing a discount code function as a “Service”
3. Adding multiple payment options (e.g. Credit Card, Debit Card, Gift Card, Cash)
4. Adding a “Reviews” page
5. Ensuring passwords are encoded using an MD5() Function prior to being saved in the database.

In order to successfully execute the creation of this OSP, we used AngularJS for the SPA implementation, HTML coding, CSS styling, JSON/Javascript scripts, PHP classes, a MySQL database through the XAMPP software, and MD5 for data security.

Multi-pages and User Interfaces

1. UI For Drag and Drop and Shopping Cart



FOOTWEAR

30% off



ADIDAS

Model: Adizero, Color: Cream-White

\$75 ~~\$107.14~~



SALOMON

Model: Salomon XT-6 GTX, Color: Black

\$160

Shopping Cart

Your cart is empty.

Total: \$0.00

[Proceed to Checkout](#)

FOOTWEAR

30% off



ADIDAS

Model: Adizero, Color: Cream-White

\$75 ~~\$107.14~~



SALOMON

Model: Salomon XT-6 GTX, Color: Black

\$160

Shopping Cart

Your cart is empty.

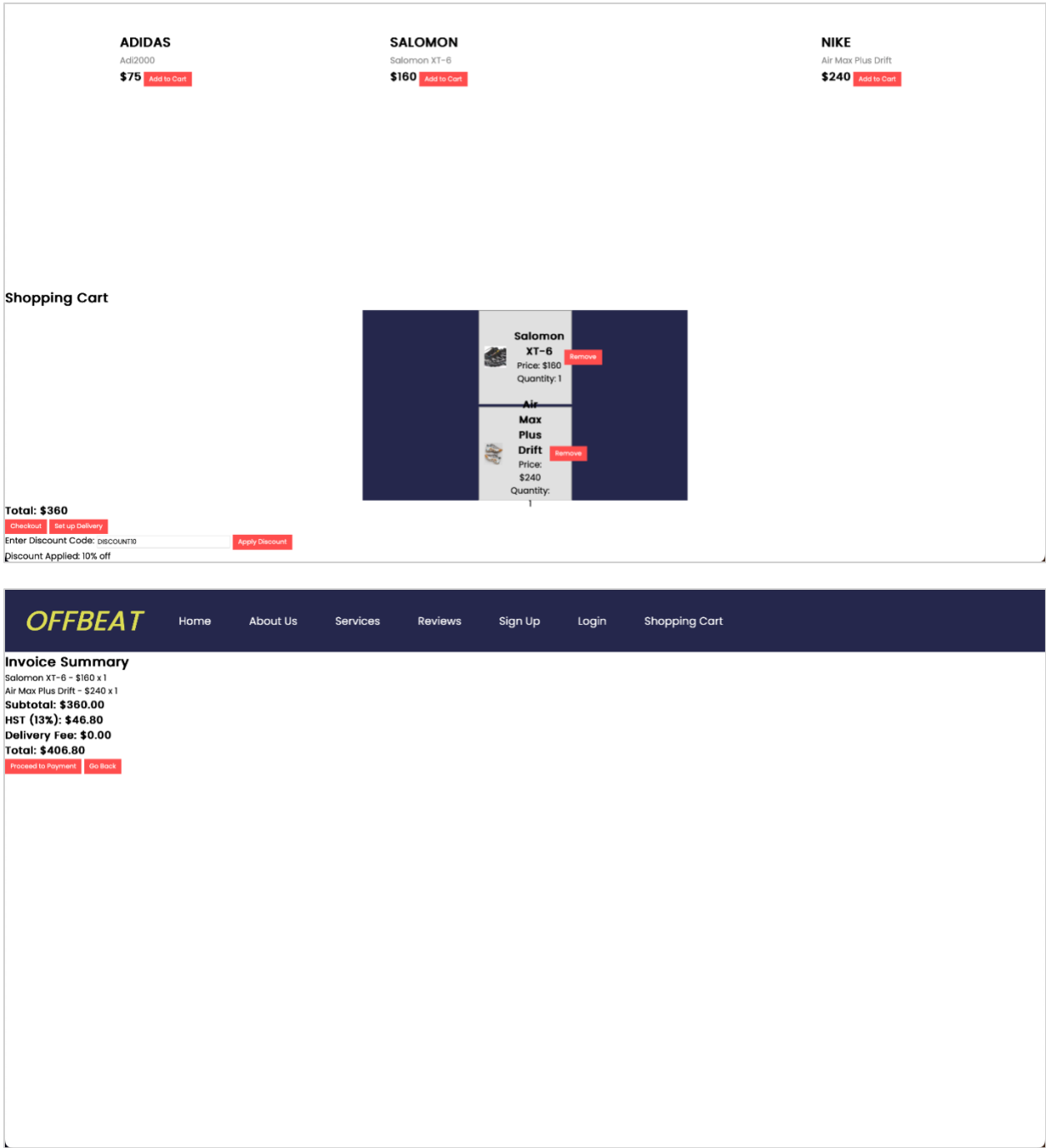
Total: \$160.00

[Proceed to Checkout](#)



1

2. UI for Checkout and Order Confirmation



OFFBEAT

[Home](#)[About Us](#)[Services](#)[Reviews](#)[Sign Up](#)[Login](#)[Shopping Cart](#)

Secure Payment

Select Payment Method

Card Number: 1234 5678

Pay Now

Back to Home

✓ Credit Card

Debit Card

Cash

Gift Card

Expiry Date: MM/YY

CVV: 123

OFFBEAT

[Home](#)[About Us](#)[Services](#)[Reviews](#)[Sign Up](#)[Login](#)[Shopping Cart](#)

Secure Payment

Select Payment Method: Debit Card

Card Number: 12345678901234

Pay Now

Back to Home

Expiry Date: 1234

CVV: 123

Payment Successful!

Close

Secure PaymentSelect Payment Method: Debit Card

Card Number: 12345678901234

Expiry Date: 1234

CVV: 123

[Pay Now](#)[Back to Home](#)

Receipt Generated: ID 13

[Close](#)

phpMyAdmin

Server: localhost Database: Project-R1-R2-W25 Table: Shopping

Recent Favorites

SELECT * FROM `Shopping`

Showing rows 0 - 12 (13 total, Query took 0.0002 seconds.)

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	Receipt_Id	Store_Code	Total_Price
  	1	STORE001	185.80
  	2	STORE001	163.20
  	3	STORE001	185.80
  	4	STORE001	270.55
  	5	STORE001	724.81
  	6	STORE123	1017.00
  	7	STORE123	94.75
  	8	STORE123	254.08
  	9	STORE123	94.75
  	10	STORE123	86.28
  	11	STORE123	391.55
  	12	STORE123	335.44
  	13	STORE123	406.80

Check all | With selected:    

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

Bookmark this SQL query

Console

Press Ctrl+Enter to execute query

```
>SELECT * FROM `Users`  
>SELECT * FROM `Customers`  
>SELECT * FROM `Reviews`
```

3. UI for Delivery

OFFBEAT [Home](#) [About Us](#) [Services](#) [Reviews](#) [Sign Up](#) [Login](#) [Shopping Cart](#)

Find the Best Delivery Route

Select Warehouse:

Toronto Warehouse

Enter Customer Address:

5 Ann Street

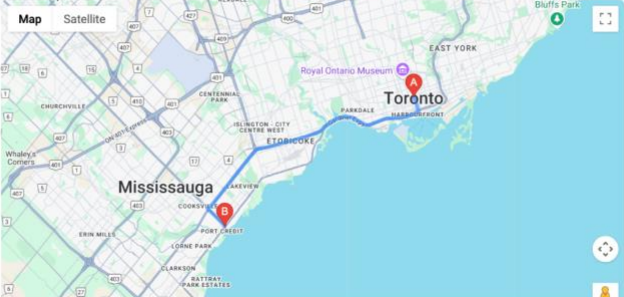
Select Delivery Date:

2025-03-04

Find Best Route

Go Back

Map Satellite



4. UI for Sign-up and Log-in

SIGN UP

Provide your info

 Firstname

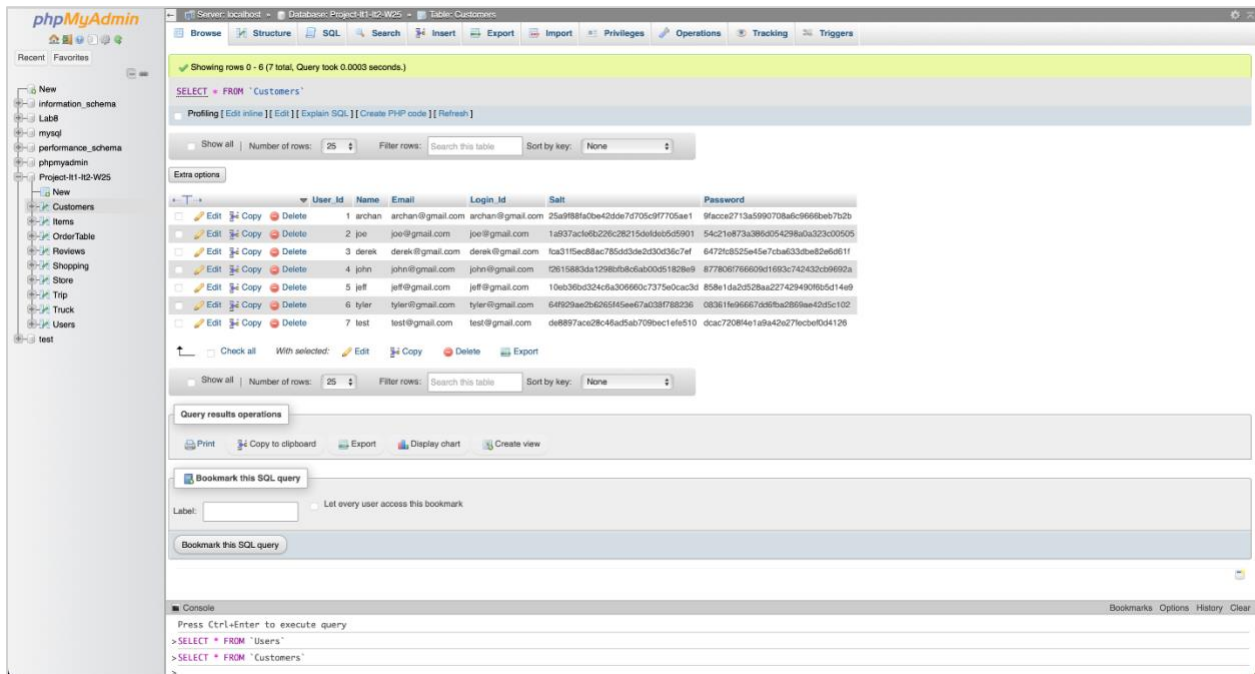
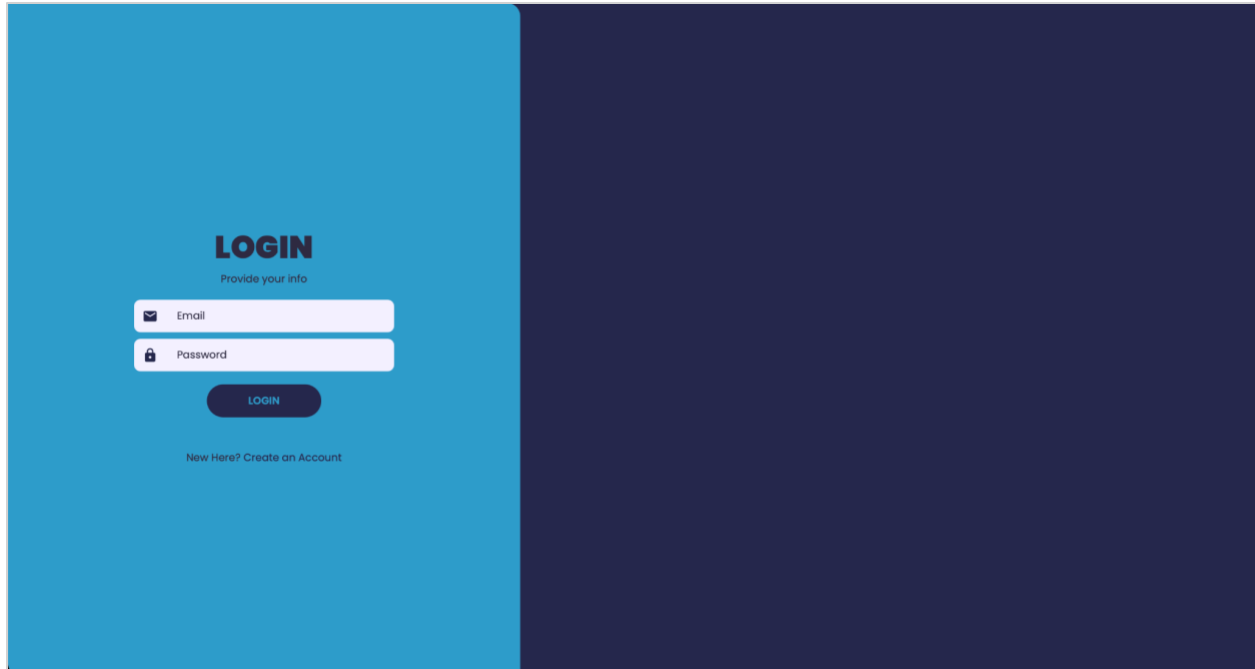
 Email

 Password

 Repeat Password

SIGN UP

Already have an Account? [login](#)



5. UI for Reviews

OFFBEAT [Home](#) [About Us](#) [Services](#) [Reviews](#) [Sign Up](#) [Login](#) [Shopping Cart](#)

Write a Review
Rate your experience (1-5): 5 - Excellent Write a short review: [Submit Review](#)

Customer Reviews:
Rating: 5 / 5
Shoes were great

phpMyAdmin

Server: localhost Database: Project-R1-R2-W25 Table: Reviews

Showing rows 0 - 0 (1 total, Query took 0.0004 seconds)

`SELECT * FROM `Reviews``

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

	id	RN	Review
☐		5	Shoes were great

Check all | With selected: ☐ Edit ☐ Copy ☐ Delete ☐ Export

Show all | Number of rows: 25 | Filter rows: Search this table

Query results operations

Print Copy to clipboard Export Display chart Create view

Bookmark this SQL query

Label: ☐ Let every user access this bookmark

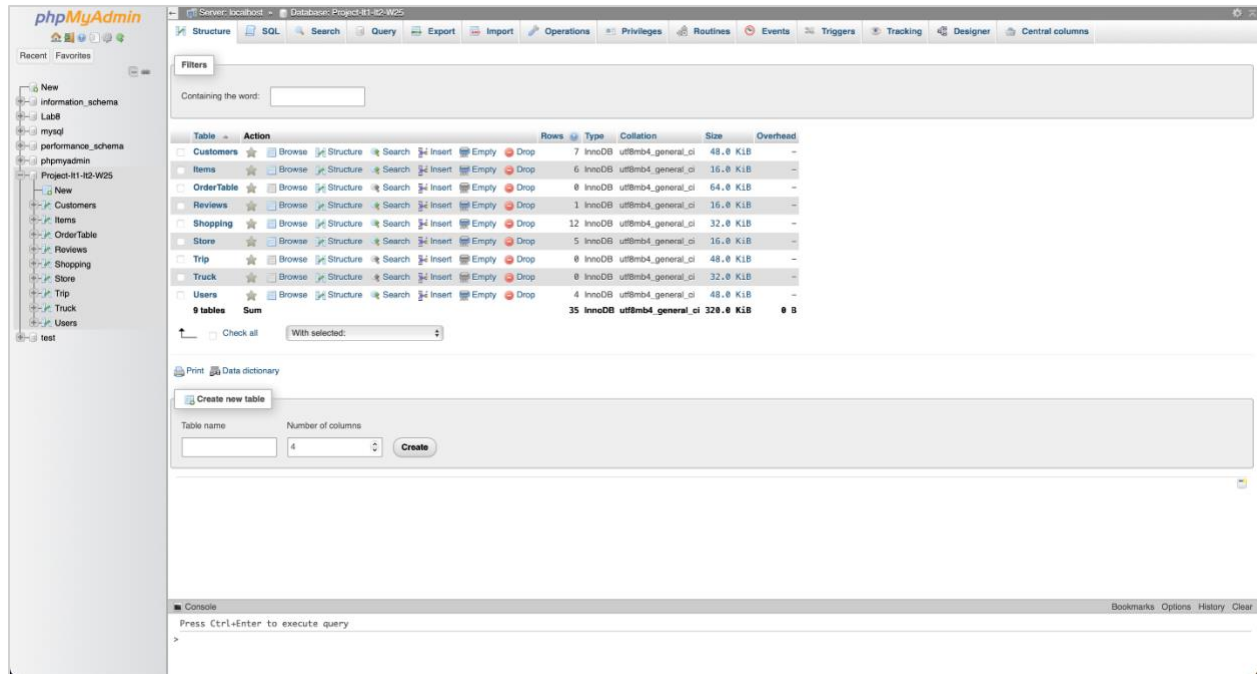
Bookmark this SQL query

Console

Press Ctrl+Enter to execute query

```
>SELECT * FROM `Users`  
>SELECT * FROM `Customers`  
>SELECT * FROM `Reviews`
```

6. Databases



MD5() Implementation and Security Risks

The primary security risk with our sign-up system was the plain text password storing. If an external source was able to access our database, they would immediately learn all users' passwords, which would be a massive liability. This was prevented by implementing the MD5 algorithm below.

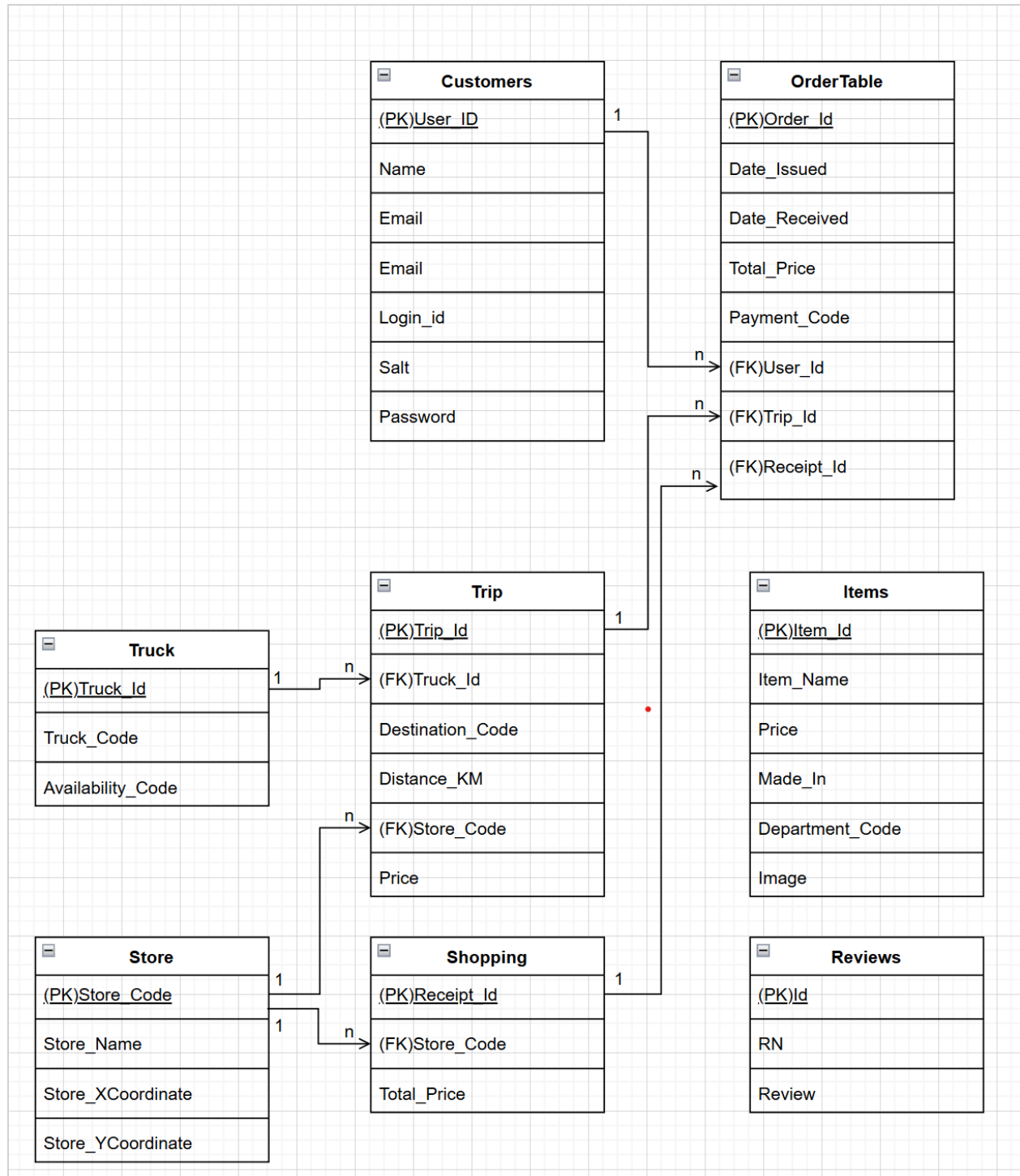
```
$salt = bin2hex(random_bytes(16)); // Generate a random 16-byte salt
```

```
$hashedPassword = md5($salt . $password); // Hash password with salt using MD5
```

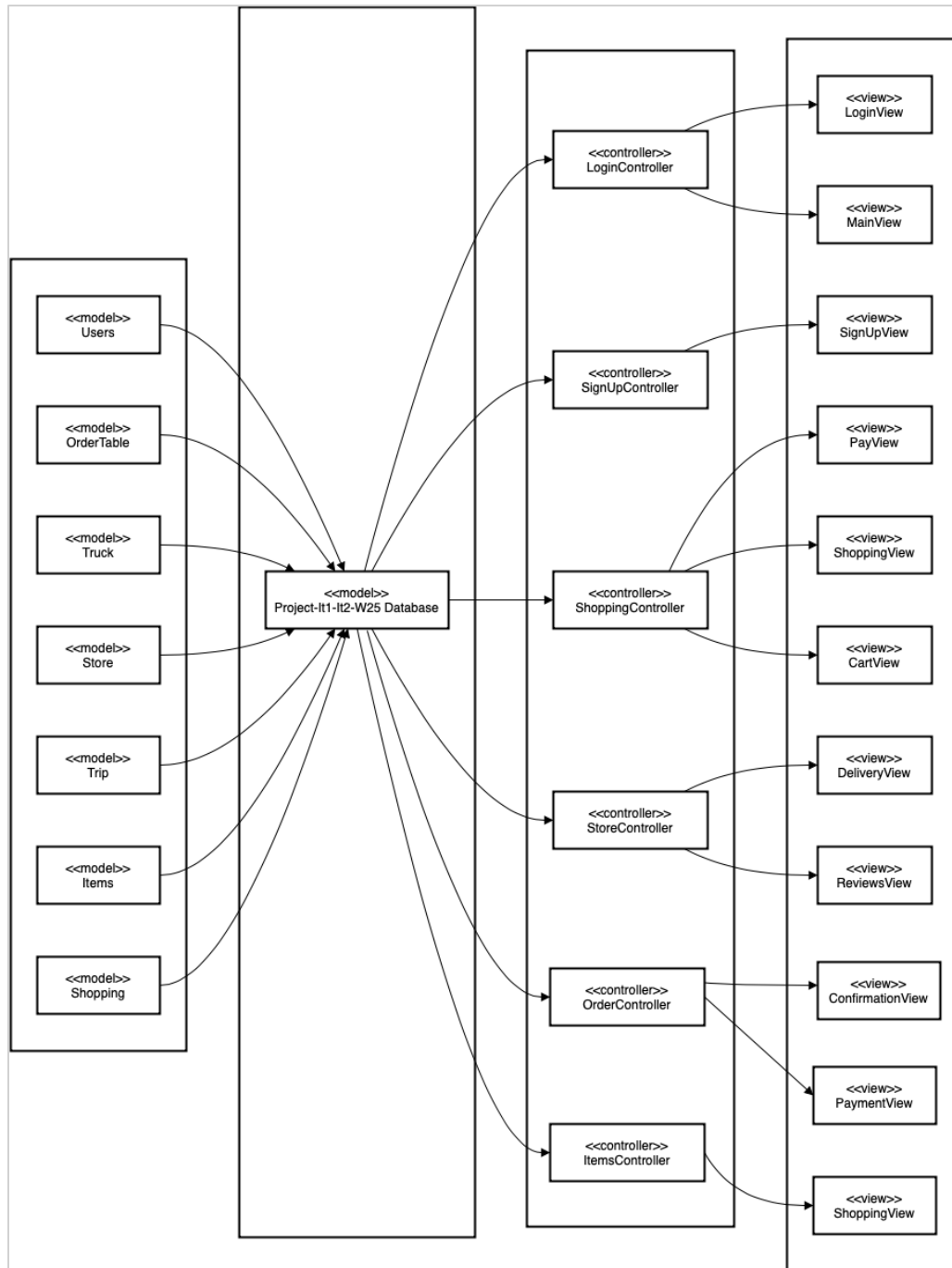
The MD5 algorithm used here will concatenate the salt variable with the user's entered password, and the hashed result will be stored in the "Customers" Database.

User_Id	Name	Email	Login_Id	Salt	Password
1	archan	archan@gmail.com	archan@gmail.com	25a9f88fa0be42dde7d705c9f7705ae1	9facce2713a5990708a6c9666beb7b2b
2	joe	joe@gmail.com	joe@gmail.com	1a937acf06b226c28215defdeb5d5901	54c21e873a386d054298a0a323c00505
3	derek	derek@gmail.com	derek@gmail.com	fca31f5ec88ac785dd3de2d30d36c7ef	6472fc8525e45e7cba633dbe82e6d61f

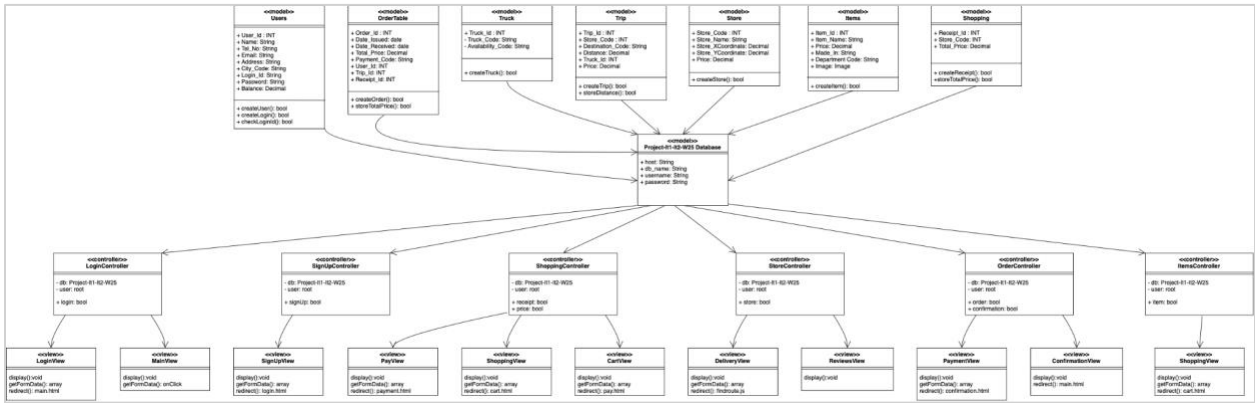
Case Study Database Schema



MVC Web-Application Architecture – General Class Diagram



MVC Web-Application Architecture - Detailed Class Diagram



SPA Architecture Diagram

